Assignment 1

Fatima Madey
MPCS 56540
April 10, 2026

Exercise 1 - State-Based Testing

Vending Machine

Consider a simplified vending machine with the following states:

- Idle
- Accepting Coins
- Dispensing Product
- Out of Stock

Assumptions

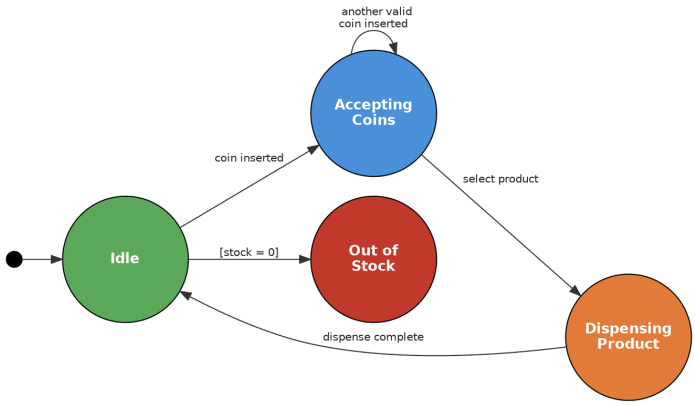
- The machine starts in the Idle state.
- When a coin is inserted, the machine moves from Idle to Accepting Coins.
  - If the machine is out of stock, it moves to Out of Stock.
- If another valid coin is inserted while already in Accepting Coins, it remains in Accepting Coins.
- If the user selects a product while in Accepting Coins, it moves to Dispensing Product.
- After dispensing, the machine returns to Idle.
- Assume there is only one product type and one valid coin type.
- Ignore change return and refund behavior.

Tasks

- Draw a diagram for the vending machine.
- Create a state transition table.
- Design 6-8 test cases that cover:
  - all valid state transitions;
  - at least two invalid or disallowed transitions;
  - expected system behavior in each relevant state.

Deliverables

Diagram



State Transitions Table

Current State	Event / Condition	Next State	Action / Output
Idle	coin inserted	Accepting Coins	Accept coin
Idle	coin inserted + out of stock	Out of Stock	Display "Out of Stock"
Accepting Coins	another valid coin inserted	Accepting Coins	Accept coin
Accepting Coins	select product	Dispensing Product	Dispense product
Dispensing Product	dispense complete	Idle	Reset for next customer
Out of Stock	(any event)	Out of Stock	Remain out of stock

Test Cases

Test ID	Test Name	Current State	Input / Action	Expected Next State	Expected Behavior
1	vaildCoinAccepted	Idle	Insert coin	Accepting Coins	Coin accepted
2	multipleCoinsAccepted	Accepting Coins	Insert another coin	Accepting Coins	Coin accepted
3	successfulSelection	Accepting Coins	Select product	Dispensing Product	Product dispensed
4	successfulPurchase	Dispensing Product	Dispense complete	Idle	Ready for next customer
5	validOutOfStock	Idle	Stock = 0	Out of Stock	"Out of Stock" message
6	invalidSelectionBeforePayment	Idle	Select product	Idle	Error / ignore (no product selected)
7	invalidInsertionDuringSelection	Dispensing	Insert Coin	Idle	Reject Coin or Store for Next Iteration
8	fullCycle	Idle	Coin → select → dispense	Idle	Complete transaction

Exercise 2 - Equivalence Partitioning and Boundary Value Analysis

Shipping Cost Calculator

You are testing a shipping cost calculator for an e-commerce website. Shipping cost is determined by package weight.

Rules

- Lightweight package: weight < 5 lb
- Medium-weight package: 5 lb ≤ weight ≤ 20 lb
- Heavy package: weight > 20 lb
- Minimum valid weight: 0 lb
- Maximum valid weight: 50 lb

Tasks

Using equivalence partitioning and boundary value analysis, design test cases for the weight field.

Your answer must include

1. valid equivalence classes
2. invalid equivalence classes
3. boundary values around:
    - minimum valid weight;
    - the 5 lb cutoff;
    - the 20 lb cutoff;
    - maximum valid weight.
4. A final reduced test set that gives good coverage.

Deliverables

Equivalence Classes

- Valid Equivalence Classes
    - Lightweight: 0 ≤ weight < 5 lb
    - Medium: 5 ≤ weight ≤ 20 lb
    - Heavy: 20 < weight ≤ 50 lb
- Invalid Equivalence Classes
    - Negative weight: (< 0 lb)
    - Over maximum: (> 50 lb)

Boundary Values

- Minimum valid: 0, 1 (if including invalid -1)
- 5 lb cutoff: 4, 5, 6
- 20 lb cutoff: 19, 20, 21
- Maximum valid: 49, 50, (if including invalid 51)

Test Case Table

Test ID	Weight (lb)	Expected Class	Expected Result
1	-1	Invalid	Error: weight must be ≥ 0
2	0	Lightweight	Light shipping rate
3	4	Lightweight	Light shipping rate
4	5	Medium	Medium shipping rate
5	6	Medium	Medium shipping rate

Test ID	Weight (lb)	Expected Class	Expected Result
6	19	Medium	Medium shipping rate
7	20	Heavy	Heavy shipping rate
8	21	Heavy	Heavy shipping rate
9	50	Heavy	Heavy shipping rate
10	51	Invalid	Error: weight cannot exceed 50 lb

Exercise 3 - Compatibility Testing Strategy

For a web application, design compatibility testing across:

- browsers: Chrome, Firefox, Safari;
- operating systems: Windows, macOS, Linux;
- devices: desktop, mobile, tablet.

Tasks

1. Create a compatibility matrix showing the configurations to consider.
2. Calculate how many configurations are required for exhaustive testing.
3. Propose a reduced testing strategy that would be realistic in practice.
4. Explain which combinations you would prioritize first and why.

Deliverables

Compatibility Matrix

Browser	Windows	macOS	Linux	Desktop	Mobile	Tablet
Chrome	✓	✓	✓	✓	✓	✓
Firefox	✓	✓	✓	✓	✓	✓
Safari	–	✓	–	✓	✓	✓

Exhaustive Configuration Count

3 browsers × 3 OS × 3 devices = 27 configurations

Prioritized Reduced Strategy (9 configs)

- Chrome + macOS + Desktop (highest usage)
- Chrome + Windows + Desktop
- Chrome + Android + Mobile
- Firefox + Windows + Desktop
- Safari + macOS + Desktop
- Safari + iOS + Mobile
- Chrome + iOS + Tablet
- Firefox + Linux + Desktop (edge case)
- Chrome + Windows + Tablet

Why these first?

- Chrome dominates ≈65% of the market, making it the top priority for testing.[1] Safari is the next most-used browser, though it's exclusive to Apple devices.[1] Desktop is the leading platform for e-commerce and should be the primary testing environment, followed by mobile as the fastest-growing second.[2] Linux is the lowest priority — it holds the smallest user share and is predominantly used alongside Firefox.[1]

Sources

1. https://www.browserstack.com/guide/understanding-browser-market-share
2. https://www.mobiloud.com/blog/what-percentage-of-internet-traffic-is-mobile

Exercise 4 - White-Box Coverage

Healthy Inc. Reimbursement Logic

Healthy Inc. is a medical insurance company that reimburses drug costs.

Rules

Two drug classes are used for reimbursement:

- Class A: 90% of the cost is reimbursed, with a minimum customer participation of $5.
  - Example: a $10 drug is reimbursed by $5.

    - Example: an $80 drug is reimbursed by $72.
- Class B: 50% of the cost is reimbursed.
- For some Class A drugs priced below $5, there is no reimbursement.

Use the following pseudocode for this exercise:

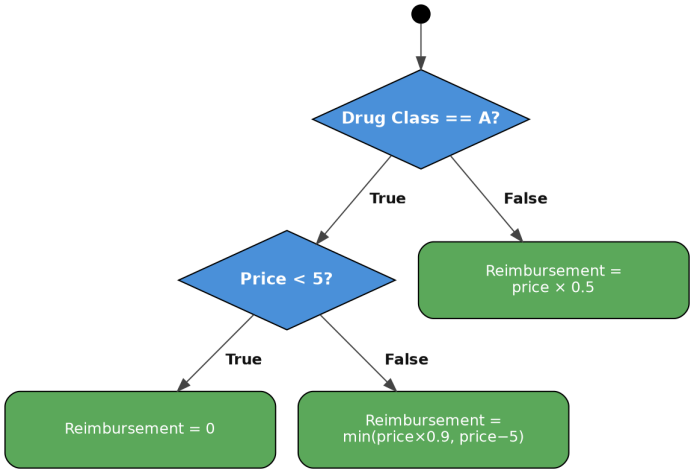
```
if drugClass == A:
    if price < 5:
        reimbursement = 0
    else:
        reimbursement = min(price * 0.9, price - 5)
elif drugClass == B:
    reimbursement = price * 0.5
```

Tasks

1. Convert the logic above into a program flow graph.
2. Provide test cases needed to achieve:
    - 100% statement coverage
    - 100% branch coverage
    - 100% path coverage
3. For each test case, map it to the parts of the graph it covers.

Deliverables

Program Flow Graph



Coverage-Oriented Test Case Table

- Statement Coverage

Test ID	Drug Class	Price	Nodes Covered	Reimbursement
1	A	3	N1, N2, N3	0
2	B	100	N1, N5	50
3	A	80	N1, N2, N4	72

- Branch Coverage

Test ID	Drug Class	Price	Branch Taken	Reimbursement
1	A	3	N1=Yes, N2=Yes	0
2	A	80	N1=Yes, N2=No	72
3	B	100	N1=No	50

- Path Coverage

Test ID	Drug Class	Price	Path Taken	Reimbursement
1	A	3	N1 → N2 → N3	0
2	A	80	N1 → N2 → N4	72
3	B	100	N1 → N5	50

Exercise 5 - Gray-Box API Testing

Gray-box testing is a testing approach that combines elements of black-box and white-box testing. In gray-box testing, the tester does not have access to the full source code, but does have some partial knowledge of the system, such as an API specification, architecture information, database schema, or workflow rules. This allows the tester to design stronger tests than pure black-box testing while still testing from an external perspective.

In this exercise, you are given access to an Order Submission API specification, but not the source code.

API Endpoint

`POST /api/v1/orders`

Headers

`Authorization: Bearer <token>` (required)

`Content-Type: application/json`

Request Body

```
{
  "customer_id": "string",
  "items": [
    {
      "product_id": "string",
      "quantity": "integer"
    }
  ],
  "delivery_date": "YYYY-MM-DD",
  "priority": "normal" | "rush"
}
```

Response Expectations

- `200 OK`: success
- `400 Bad Request`: missing or invalid fields
- `401 Unauthorized`: bad or missing token
- `422 Unprocessable Entity`: logical business issue, such as out-of-stock
- `500 Internal Server Error`

Tasks

Write 6 test cases covering:

- 2 valid requests;
- 2 invalid structure/data cases that should return `400`;
- 1 business rule failure that should return `422`;
- 1 security failure that should return `401`.

For each test case, specify:

- test name / ID;
- headers;
- input JSON;
- expected status code;
- expected response pattern;
- reason for expected result.

Assumption

For the `422` case, you may assume a documented business rule such as out-of-stock or another clear logical fulfillment failure.

Deliverables

Table of 6 Test Cases

Test ID	Test Name	Headers	Input JSON (body)	Expected Status	Expected Response Pattern	Reason (gray-box knowledg
1	Valid normal order	Authorization: Bearer `<valid-token>`	`{"customer_id":"C123","items":[{"product_id":"P1","quantity":2}],"delivery_date":"2026-04-20","priority":"normal"}`	200	`{"order_id":"...","status":"created"}`	All fields valid

Test ID	Test Name	Headers	Input JSON (body)	Expected Status	Expected Response Pattern	Reason (gray-box knowledg
2	Valid rush order	Authorization: Bearer `<valid-token>`	same as 1 but let `"priority":"rush"`	200	`{"order_id": "...", "status": "created"}`	Business rule allow rush
3	Invalid – missing customer_id	Authorization: Bearer `<valid-token>`	`{"items":[...],"delivery_date":"2026-04-20","priority":"normal"}`	400	`{"error": "missing field: customer_id"}`	Required field (API spec)
4	Invalid – type for product_id is integer	Authorization: Bearer `<valid-token>`	`{"customer_id":"C123","items":[{"product_id": 1,"quantity":1}],"delivery_date":"2026-04-20","priority":"normal"}`	400	`{"error": "quantity must be positive"}`	Type Erro
5	Business rule – out of stock	Authorization: Bearer `<valid-token>`	same as 1 but product is documented out-of-stock	422	`{"error": "out of stock"}`	Document business r
6	Security – missing token	(no Authorization header)	same as 1	401	`{"error": "unauthorized"}`	API spec requires token

Exercise 6 - Static Analysis / Linter Demo

Install and configure a linter or static analysis tool for one of the following languages:

- Python
- Java
- JavaScript
- C++ Examples include ESLint, Pylint, Flake8, PMD, Checkstyle, or Cppcheck.

Tasks

1. Install and configure one linter/static analysis tool.
2. Configure it to detect:
    - at least 5 style-related issues
    - at least 5 logic / code-quality issues
3. Prepare one or more source files that intentionally contain issues.
4. Run the tool and capture the output.

Requirements

- You must demonstrate at least 10 findings total.
- At least 8 findings must come from distinct rule types.
- Repeating the same issue many times does not satisfy the distinctness requirement.

Deliverables

Source Code File

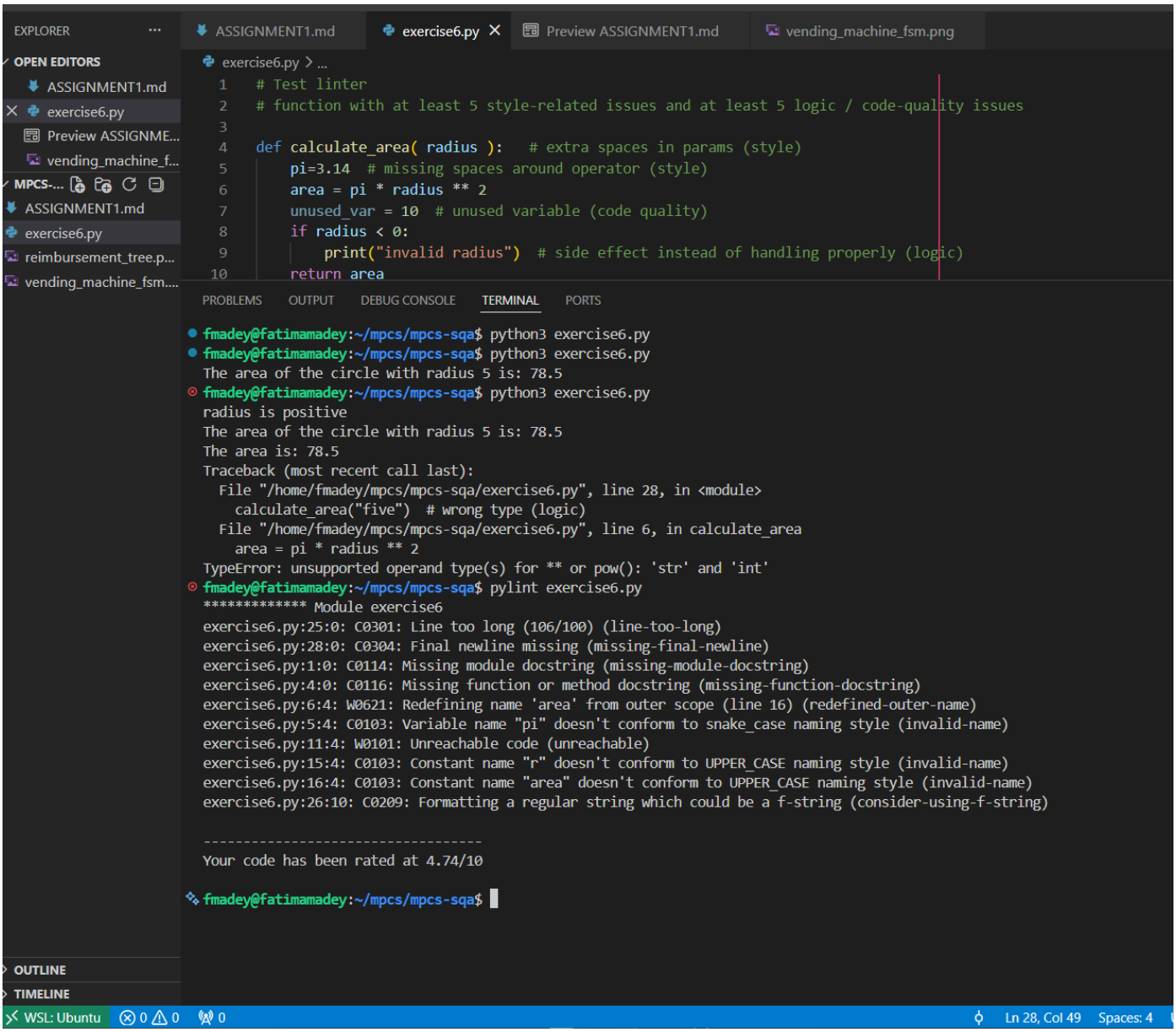
https://github.com/fatimamadey/mpcs-sqa/blob/main/assignment1/exercise6.py

Linter Configuration File

https://github.com/fatimamadey/mpcs-sqa/blob/main/assignment1/config.pylintrc

Source: https://pylint.pycqa.org/en/stable/user_guide/configuration/index.html

Screenshots of Terminal



Use of External Resources

- Used Claude AI to generate graphs based on node description and flow.
- Also used Claude to do some formatting for my markdown file.
- Used copiolet autocomplete to help me write a function that i later added linter errors to for exercies 6.
- Asked Claude to grade my assignment to catch any small errors I might have made.
  - made a few typos but logic was good (also claude is a harsh grader...)
- Claude convo (https://claude.ai/share/14393ff8-b06f-46d1-b219-079555559aa3)